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Importance of aspect ratio over shape in determining the quantization potential of self-assembled zinc-blende III-V quantum dots

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We have studied the effect of shape on the strain-modified electron/hole confinement potential in zinc-blende quantum dots (QDs), using standard deformation potential theory and an anisotropic continuum-elasticity approximation. Calculations were performed for a variety of shapes of InAs/GaAs (001) QDs. Our results show that the essential features of the confinement potential are determined primarily by one geometric parameter, i.e., the aspect ratio, being insensitive to other details of the QD shape. The underlying trends in strain distribution are also revealed. Our results suggest that a simple analytical model, based on the oblate-ellipsoid shape and isotropic elasticity approximation, adequately describes the major features of the strain-modified confinement potential for a wide range of self-assembled III-V (zinc-blende) QDs. © 2007 American Institute of Physics. [DOI: 10.1063/1.2752127]

I. INTRODUCTION

Zero-dimensional semiconductor quantum dots are commonly used now both in fundamental studies and for optoelectronic applications, such as forming the active media in semiconductor lasers. Self-assembled quantum dots (SAQDs) predominate and these are obtained by the pseudomorphic Stranski-Krastanov growth method.¹ As the method is based on the lattice mismatch between dot and substrate, strain is inherent to SAQDs. It has been shown¹ that such a misfit strain will have a dramatic effect on the band profile in the SAQDs. Indeed, for typical values of the deformation potentials in III-V systems,² the shift or splitting in the conduction or valence bands can be expected to be as large as 80 meV per 1% of strain. As a typical example, the lattice misfit in the InAs/GaAs SAQDs is 7%. Therefore, the strain-induced change in the band-edge positions is expected³ to be comparable to or even larger than the band gap in the unstrained materials.

Effects of mismatch strain on the band structure have been reported for a variety of QD shapes, such as flat cylinders,⁴ pyramids,⁵ lenses,^{3,6} and cones.⁷ Examples include a comparative study of spherical and lens-shaped dots of a particular size,⁸ and examination of the effect of the height-to-base ratio for pyramid⁹ and truncated-pyramid¹⁰ shapes. It has been a common conclusion that the dot shape (volume) has a significant effect on the electronic structure; however, trends have largely been studied within single shape “families,” such as by taking a pyramid and varying its volume or aspect ratio. Hence, the question as to what extent the QD band structure is sensitive to particular dot geometry, has remained unanswered. This question is particularly important because it is very difficult both to gauge the exact dot geometry in already grown structures (especially in capped ones) and to predict the shape when designing new structures.

The dot shape affects the electronic spectra in two ways. First, it affects the strain distribution in and around the dot and hence modifies the potential profile for both electrons and holes (the “strain” effect). Second, variation in shape affects size quantization even though some linear dimensions remain the same. Qualitatively, this can be described as a change in the effective size of confinement in a particular direction (the “size” effect). Note that such a size effect is different from the well-studied effect of simple size scaling for a dot of a fixed shape.⁵ Since the contributions from the “strain” and “size” effects may be of opposite signs,¹⁰ it is very informative to study them separately. Such an analysis is particularly effective for strain when using the continuum-elasticity approach, since the solution of the elastic problem is size independent and scalable. In the present paper, we address only the strain-induced modification of confinement potential treated in the continuum-elasticity approximation. The size-quantization energies are not considered.

It may be helpful to consider possible reasons for the variation of strain with the dot shape. First, we need to note that the misfit deformation is distributed between the dot and the surrounding matrix. The extent of this depends both on the elastic modulus for the dot and matrix materials and on the dot shape. Intuitively, one can expect that in the case of a stiff matrix and a soft QD the deformation is localized predominantly in the QD and vice versa. However, this may not always be true, as it can be illustrated by comparing two extreme shapes, a spherical dot and an extremely flat dot. For a spherical dot in the isotropic elasticity approximation, the strain within the QD is¹¹

$$e_{xx} = e_{yy} = e_{zz} = e_m(1/\gamma - 1),$$

$$\text{where } \gamma = 1 + 2 \frac{k_m(1 - 2\sigma_m)}{k_d(1 + \sigma_m)}. \quad (1)$$

Here, e_{ii} ($i=x,y,z$) represents the diagonal components of the strain tensor; e_m is the lattice mismatch; $\sigma=(3k-2\mu)/(3k+\mu)/2$ is Poisson's ratio; k and μ are the bulk and

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shear moduli, respectively; d and m are indices referring to the dot and matrix, respectively. The hydrostatic strain within the dot varies with the ratio of bulk moduli for the dot and the matrix. However, for the “opposite” shape of an extremely flat dot, the strain distribution should approach that for a quantum well (QW), with the matrix completely unstrained and the whole misfit strain concentrated within the dot. These examples illustrate that even for fixed elastic parameters for the dot and the matrix, the strain distribution between them is shape dependent.

Second, the misfit strain in the dots is distributed between different components of deformation, which have different impacts on the band edges. This distribution is also shape dependent, as can be seen for the extreme cases described above. The strain within a spherical QD is purely hydrostatic; however, for QW-like strain in an extremely flat dot, there is also an essential tetragonal component.¹² Finally, though strain is uniform in an ellipsoidal dot,¹¹ variation of strain within the dot is possible for an arbitrary shape.

The aim of the present work is to reveal the most important geometrical parameter(s) for a QD of an arbitrary shape. [The example above already indicates that the dot aspect ratio (AR) is one geometrical parameter of importance, with the sphere having AR=1, while for a QW, AR=infinity.] We examine the InAs/GaAs material combination, which serves here as the most well-known test system. We start by taking the ellipsoidal shape as an example since this has the simplest strain distribution. We analyze the most important trends in the strain distribution and the strain-modified confinement potential as a function of the aspect ratio, which is the only variable shape parameter for an oblate ellipsoid. Next, we analyze the same trends for a number of more complicated QD shapes, as well as taking into account the effect of the wetting layer. Our results show that for a number of realistic QD shapes, the basic features of the strain and the band profile are determined by the aspect ratio only and are much less sensitive to other shape details. This allows us to justify approximating the QD shape to an ellipsoid and to suggest a simplified model for analysis of the strain-modified band profile in the SAQDs. Based on the analytical solution of the elastic problem, this model provides a straightforward and relatively accurate description of the confinement potential in SAQDs.

The paper is organized as follows. Section II formulates the problem in terms of standard deformation potential theory and describes analytical and numerical approaches to the elastic problem. The calculated results are presented and discussed in Sec. III. Section IV introduces the ellipsoidal model. It is followed by a summary in Sec. V. Details of formulae used in the calculations are provided in the Appendix.

II. METHODOLOGY

To calculate the strain-modified confinement potential in the dots and the surrounding matrix, we have applied standard deformation potential theory. A prerequisite to this is obtaining the misfit strain distribution, which was routinely modeled numerically. This was complemented by analytical

modeling of a simplified elastic problem. The elastic energy density was used to achieve a common baseline for comparative analysis of different strain components in both the dots and the matrix.

The material parameters were taken from Ref. 2. Note that all band energies in our paper are given with respect to the valence band edge in unstrained InSb, as in Ref. 2.

A. Numerical modeling of the elastic problem

The strain distribution in and around the QD for an arbitrary dot shape can only be obtained by numerical modeling. To do this, we employed a standard finite element method (FEM) using the COMSOL software package. We used the continuum-elasticity approximation and both the cubic anisotropy and the different elastic constants of the dot and matrix materials were taken into account.

Note that comparison of the continuum-elasticity approach to a more realistic atomistic-elasticity approximation has revealed¹³ a good agreement between both methods for strain calculation in QDs. In particular, discrepancies were observed only within one interatomic distance of the heterointerface, which is significant only for extremely small QDs.

B. Analytical approach to the elastic problem

For ellipsoidal QDs, the elastic problem of an ellipsoidal inclusion has a simple algebraic solution in the isotropic elasticity approximation by Eshelby.¹¹ The relevance of an ellipsoidal approximation for the QD shape will be discussed in subsequent sections. At this moment, we would like to point out two important properties of the ellipsoidal shape: (i) The algebraic solution provides the strain distribution both inside the dot and at the heterointerface, which is of key importance for analyzing the quantization potential. Thus, a broad range of material combinations can readily be analyzed without extensive numerical calculations. (ii) Within an ellipsoidal inclusion, strain is homogeneous.¹¹ Therefore, modeling provides simple single-valued results for the confinement potential. For other dot shapes, strain varies across the dot, and only the effective potential obtained from averaging over the QD volume can be used for a simple estimate of confinement.

The relevant expressions¹¹ for six components of the strain tensor for an oblate ellipsoid are compiled in the Appendix. These expressions provide a basis for strain calculations within the proposed isotropic ellipsoidal (IE) model. A note is needed for how the cubic elastic constants for the matrix and dot materials are incorporated in the isotropic elasticity approximation. In the latter case, there are only two independent elastic constants (bulk and shear moduli, or Lamé coefficients), instead of three for cubic anisotropy. For cubic symmetry, the Hook's law can be presented in following form:^{14,15}

$$\begin{aligned}\sigma_{ij} &= ke + 2\mu_1' e_{ij} \quad (i = j) \\ &= 2\mu_2' e_{ij} \quad (i \neq j),\end{aligned}\tag{2}$$

where

$$k = (c_{11} + 2c_{12})/3, \quad \mu_1 = (c_{11} - c_{12})/2, \quad \text{and} \quad \mu_2 = c_{44}.$$

Here k and $\mu_{1,2}$ are bulk and shear moduli, respectively; $e \equiv e_{mm} = e_{xx} + e_{yy} + e_{zz}$ and $e_{ij} \equiv e_{ij} - (1/3)e\delta_{ij}$ are dilatational and deviatoric parts of the strain tensor, respectively; c_{ij} are three independent stiffness constants for cubic symmetry crystals. For the (001) growth plane, the $i \neq j$ components of the strain tensor are identically zero¹¹ within an ellipsoidal QD, i.e., deformation is purely tetragonal. Hence, k and μ_1 are a natural choice for two independent elastic coefficients. For the (111) growth plane the deformation is purely trigonal, hence μ_2 replaces μ_1 . However, for any other growth direction, only the bulk modulus stands but the shear moduli are not separable. A similar difficulty exists for the matrix material, where all the strain components are generally non-zero and an approximation for the shear modulus is needed. In the present calculations, an averaging¹⁶ of shear modulus in the matrix was employed as follows:

$$\mu_{av} = \frac{1}{5}(c_{11} - c_{12}) + \frac{3}{5}c_{44}. \quad (3)$$

C. Elastic strain energy

To characterize the strain variation with QD shape, which also takes into account the strain redistribution between the matrix and the dot, we calculate the elastic energy density. Elastic (strain) energy can be expressed as

$$\begin{aligned} F = & \frac{1}{2}c_{11}(e_{xx}^2 + e_{yy}^2 + e_{zz}^2) + c_{12}(e_{xx}e_{yy} + e_{xx}e_{zz} + e_{yy}e_{zz}) \\ & + 2c_{44}(e_{xy}^2 + e_{xz}^2 + e_{yz}^2) \\ = & \frac{1}{2}[ke_A^2 + \mu_1(e_{E1}^2 + e_{E2}^2) + \mu_2(e_{T1}^2 + e_{T2}^2 + e_{T3}^2)], \end{aligned} \quad (4)$$

The first equality is a standard form¹⁴ for the elastic free energy F . The second equality presents the same in a diagonal form. The latter can be obtained from symmetry arguments on the basis of irreducible representations of the cubic group¹⁷ or directly from Eq. (4) and a definition of F as F

$= 1/2\sigma_{ij}e_{ij}$. The diagonal representation is more convenient for the present analysis because it uses the same symmetric strain components as the deformation potential theory (see below), which are $e_A = e_{xx} + e_{yy} + e_{zz}$ (hydrostatic strain component), $e_{E1} = e_{xx} - e_{yy}$ and $e_{E2} = (1/\sqrt{3})(2e_{zz} - e_{xx} - e_{yy})$ (orthorhombic and tetragonal shear strain components, respectively), and $e_{T1} = 2e_{xy}$, $e_{T2} = 2e_{xz}$, and $e_{T3} = 2e_{yz}$ [rhombohedral (trigonal) shear strain component].

D. Effects of strain on the band profile

Once the strain components are obtained, the band profile in the strained material can be obtained as eigenvalues of a Hamiltonian composed in the additive approximation,¹⁸

$$H_{\text{total}} = H_0 + H^S + H^{\text{SO}}, \quad (5)$$

where H_0 is the Hamiltonian in absence of strain and spin-orbit coupling; H^S and H^{SO} are terms describing strain and spin-orbit coupling effects, respectively. The “strain” Hamiltonian H^S for the j -th band at $\mathbf{k}=0$ can be written as^{18,19}

$$\begin{aligned} H_j^S = & a_j(e_{xx} + e_{yy} + e_{zz}) - 3b_j[(L_X^2 - \frac{1}{3}L^2)e_{zz} + \text{c.p.}] \\ & - 6/\sqrt{3}d_j[\{L_X, L_Y\}e_{xy} + \text{c.p.}]. \end{aligned} \quad (6)$$

Here, L is the angular momentum operator, $\{L_X, L_Y\} = 1/2(L_X L_Y + L_Y L_X)$, a_j is the hydrostatic deformation potential, b_j and d_j are shear deformation potentials for the j th band, and c.p. denotes cyclic permutations with respect to the indices x , y , and z .

The nondegenerate Γ_c conduction band responds to the lattice deformation only through the hydrostatic component of strain,

$$H_C^S = a_C(e_{xx} + e_{yy} + e_{zz}), \quad (7)$$

where a_C is the hydrostatic deformation potential.

For the valence band, effect of strain can be expressed in terms of the hydrostatic, shear tetragonal, and shear rhombohedral strain components in the representation from Ref. 18,

$$H_V^S = a_V(e_{xx} + e_{yy} + e_{zz}) + b \cdot \begin{vmatrix} 2e_{xx} - (e_{yy} + e_{zz}) & 0 & 0 \\ 0 & 2e_{yy} - (e_{xx} + e_{zz}) & 0 \\ 0 & 0 & 2e_{zz} - (e_{xx} + e_{yy}) \end{vmatrix} + \sqrt{3}d \cdot \begin{vmatrix} 0 & e_{xy} & e_{xz} \\ e_{xy} & 0 & e_{yz} \\ e_{xz} & e_{yz} & 0 \end{vmatrix}. \quad (8)$$

Here, a_V is the hydrostatic deformation potential, and b and d are shear deformation potentials for the valence band.

The spin-orbit Hamiltonian for the valence band in representation $(x\uparrow, y\uparrow, z\uparrow, x\downarrow, y\downarrow, z\downarrow)$ is

$$H^{\text{SO}} = \frac{1}{3}\Delta \cdot \begin{vmatrix} 0 & -i & 0 & 0 & 0 & 1 \\ i & 0 & 0 & 0 & 0 & -i \\ 0 & 0 & 0 & -1 & i & 0 \\ 0 & 0 & -1 & 0 & i & 0 \\ 0 & 0 & -i & -i & 0 & 0 \\ 1 & i & 0 & 0 & 0 & 0 \end{vmatrix}, \quad (9)$$

where Δ is the spin-orbit splitting energy [Ref. 18].

In the above Hamiltonian, we neglect the effect of strain on the spin-orbit interaction. The effect of piezoelectric field is also not included. This field was shown¹ to have a minor effect on quantization potential for SAQDs of realistic size for the most popular (001) growth plane. In addition, we need to note that the piezoelectric charge distribution is not scalable and hence it should be recalculated for any particular QD size and shape.

Within the analytical approach, strain-induced shifts/splitting of band edges can still be obtained from the general form of Eq. (5). However, we would like to note that for the oblate ellipsoid these calculations are straightforward even for the valence band due to homogeneous hydrostatic plus tetragonal ($e_{xx}=e_{yy}$) strain for the (001) plane [or hydrostatic plus trigonal strain for the (111) plane]. The relevant expressions¹⁸ for the heavy- and light-hole potentials are identical to those for quantum wells for the (001) growth plane,

$$\begin{aligned} E_{hh} &= E_V^0 + a_V \text{Tr}(e) + \frac{1}{3}[\Delta + \delta(e)], \\ E_{lh} &= E_V^0 + a_V \text{Tr}(e) - \frac{1}{6}[\Delta + \delta(e)] \\ &\quad + \frac{1}{2}\sqrt{[\Delta + \delta(e)]^2 - \frac{8}{3}\Delta\delta(e)}, \end{aligned} \quad (10)$$

where Δ is the spin-orbit splitting and $\delta(e) = -3b[e_{zz} - (e_{xx} + e_{yy})/2]$. For the (111) growth plane, similar equations are valid, with $\delta(e) = -3\sqrt{3}de_{xy}$.¹⁸

III. CALCULATION RESULTS AND DISCUSSION

A. Ellipsoidal quantum dots

In this section, we examine general trends in the strain distribution and the band profile for ellipsoidal QDs. Both strain and the strain-modified quantization potentials are calculated for the oblate ellipsoidal QD as functions of the dot aspect ratio. The most important strain-induced features of the potential and their variation with aspect ratio are revealed. The exact numerical solution for the elastic problem is analyzed in parallel to the analytical solution in the isotropic approximation, to reveal the effect of elastic anisotropy.

1. Inside the dot

We start with analysis of strain inside the dot, in terms of the elastic energy density. Figure 1(a) shows the spatial distribution of the total strain energy density for several aspect ratios obtained from numerical modeling. It presents two-dimensional cross sections for two representative planes, (010) and (-110) on the left and right halves of each picture, respectively. First of all, we note that coloration inside the dot is flat, which indicates homogeneous strain. This illustrates that for cubic anisotropy, the condition of homogeneous strain distribution within an ellipsoidal dot is still valid.^{11,20} With increasing aspect ratio, subtle change in the coloration intensity inside the dot can be observed, indicating a slight increase in the total elastic energy density.

As we have mentioned earlier, strain inside an ellipsoidal (001) dot can be represented as the sum of hydrostatic and tetragonal components. Figure 1(b) shows spatial distribution of the hydrostatic component of the strain. As expected, hy-

drostatic strain inside the dot is also homogeneous. However, variation in the coloration intensity with aspect ratio is much stronger. Together with the subtle variation in total strain, this is indicative of redistribution of strain between the two components.

Homogeneous strain inside the dot allows characterization using accurate single values. Figure 2(a) shows both the total strain energy density and its components as a function of the aspect ratio. Circles represent results obtained from numerical modeling. One can see that the hydrostatic component strongly decreases and the tetragonal component increases with AR. Such behavior is in sharp contrast to the weak variation in the total strain. The latter indicates that there is no essential redistribution of misfit strain between the dot and the matrix. Thus, redistribution of strain between the hydrostatic and tetragonal components is a prevailing manifestation of variation in strain inside the dot with aspect ratio for the (001) growth plane.

Homogeneous strain inside the dot also results in well-defined positions of the edges of the conduction and valence bands. Their values, obtained from numerical calculations, are shown by circles in Fig. 2(b). In agreement with Eq. (7), the conduction band edge shifts downwards with the aspect ratio, following the hydrostatic strain. As to the valence band, the major effect comes from increasing splitting induced by the shear strain component. Indeed, shear deformation potentials b and d , Eq. (8), are typically two to five times larger than the hydrostatic potential a_V . Hence, the top valence band shifts upwards. As a result, the band gap inside the dot drops from ≈ 1.05 eV for AR=1 to ≈ 0.6 eV for AR=20, with almost 60% of the effect originating from the shift of the valence band.

In the limit of infinite aspect ratio, both the strain and the band-edge positions approach the quantum-well values, which are shown by short solid bars in the right-hand side of Figs. 2(a) and 2(b), respectively. Note, however, that even for AR=20, the difference between the quantum-dot and quantum-well values is still large.

In order to analyze the effect of real cubic elastic anisotropy, the same calculations for ellipsoidal QDs were performed analytically in the isotropic elastic approximation (IE model). The results are shown in Figs. 2(a) and 2(b) by solid lines. In general, a good agreement is observed between isotropic and anisotropic elasticity approaches. In particular, the difference is negligible for the valence band edge; there is a minor systematic overestimation for the conduction band-edge position in the isotropic model. The underlying variations in the strain components are also well reproduced in the isotropic approximation, both qualitatively and quantitatively.

2. Outside the dot

The aspect ratio of the dot also affects the strain distribution in the matrix near the heterointerface. One can see from Fig. 1(a) that the strain energy density in the matrix is distributed nearly homogeneously around the spherical dot, being weakly modulated by elastic anisotropy. When the

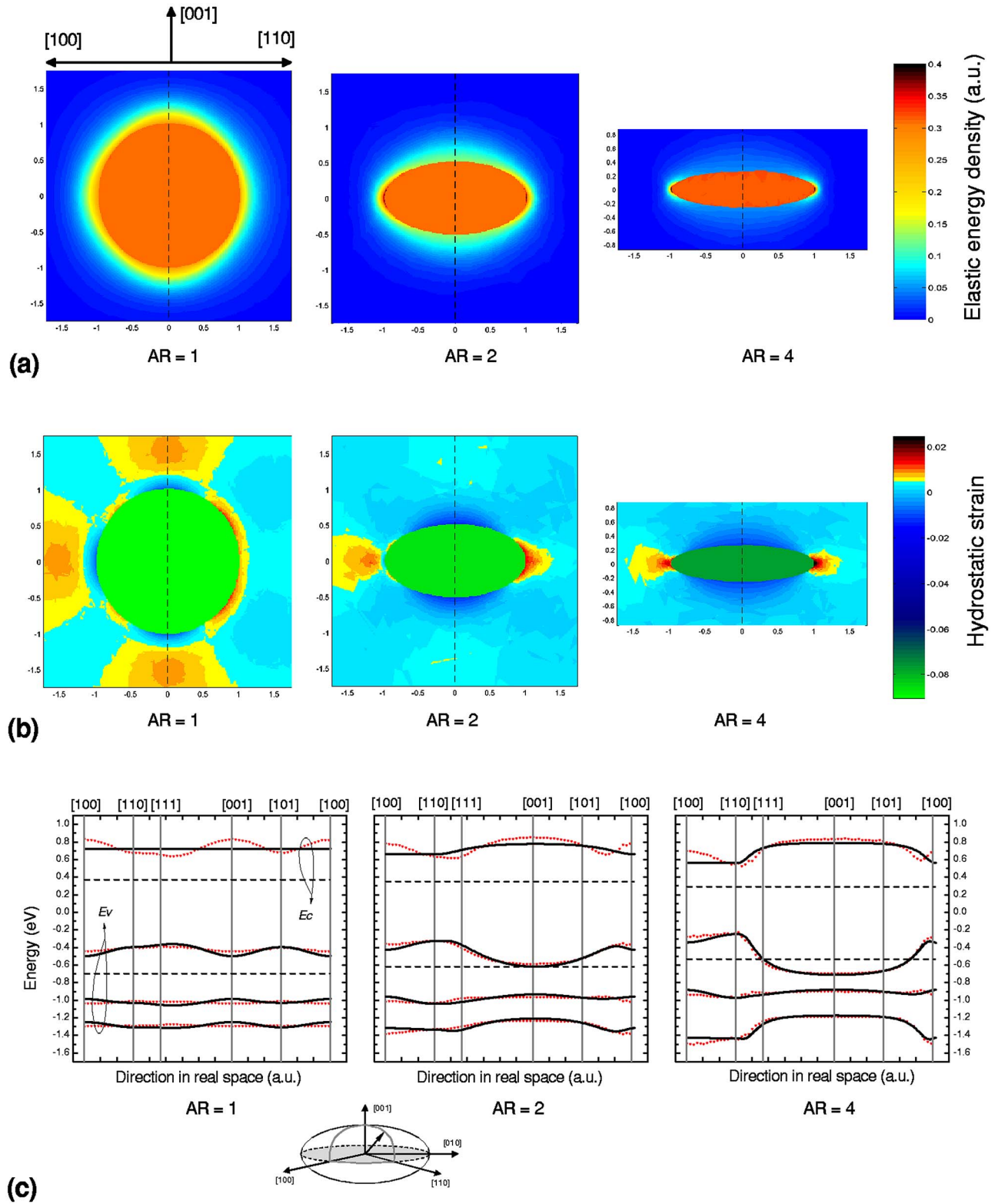


FIG. 1. (Color online) (a) Spatial distribution of total strain energy density inside and around InAs/GaAs (001) ellipsoidal quantum dot for three aspect ratios. Each graph shows (100) and (-110) cross-section planes through the dot center on the left- and right-hand-side parts, respectively. (b) As in (a) but for the hydrostatic strain component. (c) The band-edge profiles in the matrix at the heterointerface for different aspect ratios: symbols—numerical results obtained for cubic elastic anisotropy; solid lines—calculated within the isotropic elasticity approximation (IE model). Dashed lines show the band-edge positions in QD. Insert picture shows the contour at the heterointerface, along which the band profiles are presented.

aspect ratio of an ellipsoid increases, the strain is concentrated near the sharp edge of dot, while the average strain density energy along the flattened part of the interface decreases. Such behavior is in agreement with the unstrained

matrix condition for infinite AR (quantum-well limit). Note that similar behavior is demonstrated by the shear strain (not shown), which provides a dominant contribution to the total strain in the matrix.

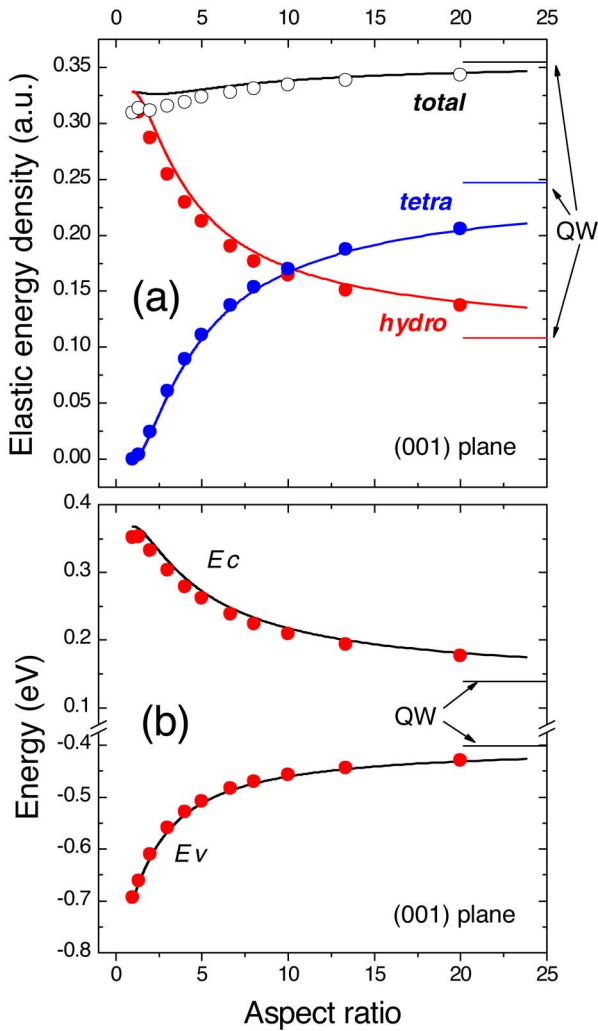


FIG. 2. (Color online) (a) Total elastic energy density and its hydrostatic and tetragonal components inside InAs/GaAs (001) ellipsoidal quantum dot as a function of aspect ratio AR. (b) Positions of conduction and top valence band edges inside InAs/GaAs (001) ellipsoidal QD as a function of AR. Symbols are numerical results obtained within cubic elastic anisotropy model; solid lines are analytical results obtained within isotropic elasticity model; short solid bars in the right-hand side of the plots are values for InAs/GaAs (001) quantum well.

At the same time, variation of the relatively weak hydrostatic strain component with AR is more complicated. One can see from Fig. 1(b) that even for a spherical dot, areas of hydrostatic contraction and hydrostatic dilation alternate along the heterointerface. Such modulation is a clear manifestation of elastic anisotropy. With increasing AR, areas of hydrostatic contraction extend over the flattened parts of the heterointerface, whereas hydrostatic dilation is concentrated near the sharp edge of the ellipsoid.

Such a distribution of shear and hydrostatic strain components results in a complex modification of the band profiles. Figure 1(c) shows the profiles of the valence and conduction bands in the *matrix at the interface with the dot* for three aspect ratios. It presents the positions of the conduction and three valence band edges along a representative contour [100]–[110]–[001]–[100] [which is marked by a bold line in the insert of Fig. 1(c)]. Dotted lines represent numerical results obtained in the anisotropic approximation, while solid

lines show analytical results in the isotropic elasticity model. Dashed lines show positions of the conduction and top valence band edges inside the dot.

Results in Fig. 1(c) clearly reveal effects from cubic anisotropy, in comparison to the IE model. For the spherical dot (AR=1), the conduction band-edge position oscillates around its value for the unstrained material. Such a behavior reflects modulation in hydrostatic strain around the interface. As mentioned above, it is a pure manifestation of elastic anisotropy. In contrast, the IE model suggests flat distribution of the conduction band edge along the interface, which reflects the zero hydrostatic strain produced by a spherical inclusion in the isotropic elastic matrix. For the valence band, a weakly varied strong splitting of the valence band is well reproduced in both elastic approximations for the same spherical shape. This splitting reflects a combination of a large, nearly homogeneous, shear strain at the heterointerface and a weak anisotropy of shear deformation potential (see also discussion in Sec. IV B below).

As the dot flattens (AR=2 and AR=4), the strain-induced valence band splitting concentrates near the sharp planar edge ([100]–[110] segment in Fig. 1(c)). This is in agreement with the strain redistribution shown in Fig. 1(a). The splitting is also well reproduced in the IE approximation. The same part of the heterointerface is also characterized by a downshift of the conduction band edge. This is due to dilation [see also Fig. 1(b)], which is reproduced in both approximations, but to a different extent. At the flattened parts of the heterointerface, small splitting in the valence band indicates low residual shear strain, while moderate uplift of the conduction band reflects the buildup of compressive hydrostatic strain. One can see that there is a very good agreement between the two models for the flattened parts of heterointerface around the [001] direction.

The above results show that major discrepancies between the isotropic and anisotropic elasticity approximations are observed only for the conduction band in the matrix. Moreover, these discrepancies are not important in determining the electronic properties of the quantum dot. Essentially, at low aspect ratios, cubic anisotropy results in directional modulation of the depth of the electron confinement potential. From this we can conclude that the flat conduction band profile in the isotropic model provides a good approximation for the mean confinement potential. This is not going to have a strong effect on the dot energy spectra, because in a spherical dot, all directions contribute equally to size quantization. For flatter dots with larger AR, the discrepancy is limited to the depth of the confinement potential in the in-plane directions. These directions provide the least contribution to the size-quantization energy. Therefore, the energy spectra of the dot cannot be strongly affected. Further analysis (below) shows that the effect becomes even weaker if the wetting layer is taken into account.

Finally, we note that the overall agreement between isotropic and anisotropic approximations is good. This suggests that, although the above analysis was performed for the (001) growth plane, a qualitatively similar modification of band profile is expected for an arbitrary growth direction.

B. Realistic QD shapes

The above analysis revealed some features of the strain-modified band profile in ellipsoidal QDs, as well as trends in its modification with the aspect ratio. To find whether they are specific for an ellipsoid, we performed a series of numerical calculations for a number of realistic QD shapes. These include a rather “smooth” hemispherical lens (dome) shape, as well as a highly “anisotropic” square-based pyramid. The results of band profile calculations are presented in Figs. 3(a) and 3(b) for the conduction and top valence bands, respectively. Cross sections for the same representative planes, (010) and ($\bar{1}10$), are shown. Geometry details for the lens, pyramid, and truncated-cone shapes are given in the insert of Fig. 3(a). Results for the ellipsoidal shape are also shown for comparison.

1. Inside the dot

For nonellipsoidal shapes, the potential profile for both electrons and holes is inhomogeneous across the dot. One can also notice that the profile becomes more homogeneous at larger aspect ratios. For dots of *all* shapes, the energy of the conduction band edge decreases and that of the valence band edge increases with the aspect ratio. These trends are indicated by variation of the overall brightness of the dot image, as shown in Fig. 3.

For quantitative comparison, energy values of the band edges are averaged over the QD volume. These *averaged* values are shown in Fig. 4(a) as a function of the aspect ratio. Symbols represent results for various shapes, obtained in the cubic-anisotropy approximation. Analytical results for the isotropic ellipsoidal model are also included. A striking observation from this figure is that all values are close to each other for a given aspect ratio. This means that the band-edge energies are not so sensitive to shape details. The most prominent difference of ≈ 0.1 eV is found for the lowest aspect ratio. This value should be considered against the strain-induced increase in the band gap up to ≈ 1 eV from 0.42 eV in the unstrained material. At higher AR, the differences become much smaller; this correlates with more homogeneous band profiles inside the dots. Also, there is a very good agreement between the isotropic and anisotropic models.

Error bars, which are shown in Fig. 4(a) for lens-shaped QDs, represent the typical mean deviation in the band-edge values. Due to better homogeneity, the deviation is smaller for larger AR. However, even for smaller AR, the mean deviation is relatively low compared to the effect produced by the variation of aspect ratio.

To reveal underlying variation of strain inside the QDs, averaged values of strain energy density components are presented in Fig. 4(b) for all the shapes as a function of the aspect ratio. One can see that the major features are mostly the same as for the ellipsoidal QD in Fig. 2(a). In particular, hydrostatic and tetragonal strains are the dominant components except for very low AR. Enhancement in the tetragonal component at the expense of the hydrostatic one is also a major effect of the increased AR. At lowest aspect ratios, however, the trigonal and orthorhombic components are the

most prominent shear components of the strain energy. (The orthorhombic component is not shown because it contributes $\sim 10\%$ at most²¹ to the total energy density.)

Taking into account ambiguity in the definition of the aspect ratio, agreement between the results for different shapes is remarkable, both in terms of strain and band profile. This justifies the role of the aspect ratio as a governing parameter related to the QD shape. Note that the importance of the aspect ratio correlates with an earlier suggestion¹⁰ regarding the significance of such QD parameter as a volume.

We note that our results are in good qualitative agreement with those reported in the literature, taking into account deviations in materials parameters, elastic constants, dot shapes, and/or the approximations used. A quantitative comparison with Ref. 3 is possible, where similar material parameters and averaging procedures were used for lens-shaped QDs of $AR \approx 3$. Relevant averaged band-edge values are represented by filled symbols in Fig. 4(a). They are in excellent agreement with our results.

2. Outside the dot

Analysis of band alignment in the matrix also reveals features which are common for different QD shapes. This can be seen from Fig. 3. In addition, representative one-dimensional (1D) band profiles along the [001] direction are provided in Fig. 5. In each graph, band profiles for dots of the same size and aspect ratio are shown, which allows easy comparison.

At low aspect ratio, the strain effect on the conduction band edge can be described qualitatively as uplift [darker areas in Fig. 3(a)] at some regions of the heterointerface, which coexists with downshift (lighter areas) of comparable magnitude in other regions. (This behavior has been reproduced in earlier calculations for different QD shapes.^{1–9}) Accurate potential profile is sensitive to particular shape details, as one can see from Fig. 5(a). It is worth noting, however, that because of alternating areas of uplift and downshift, a 1D band profile is also direction specific. As a result, a 1D band profile along the [001] direction alone is not sufficient for representing the effective localization potential.

Modification of the conduction band edge with aspect ratio in Fig. 3(a) displays a similar trend for all shapes. It can be described as the development of uplift (darker areas) near the flattened sides of the QD accompanied by downshift (lighter areas) along the sharp in-plane corners. It is notable that the uplift is observed even for the pyramidal shape, despite its sharp tip! Such a band profile matches the hydrostatic strain profile depicted in Fig. 1(b) for the ellipsoidal shape. Moreover, the magnitude of the uplift is also similar for different dot shapes, which is demonstrated by the 1D band profiles along the [001] direction in Fig. 5(b). The only substantial difference is observed at the pyramid tip. However, this effect is very local, as is illustrated in Fig. 5(c) by the band profile along the [001] line which is slightly shifted from the dot center.

For a flat QD, the [001] direction produces the dominant contribution to the size quantization of electrons. Hence, the one-dimensional conduction band profile in Figs. 5(b) and

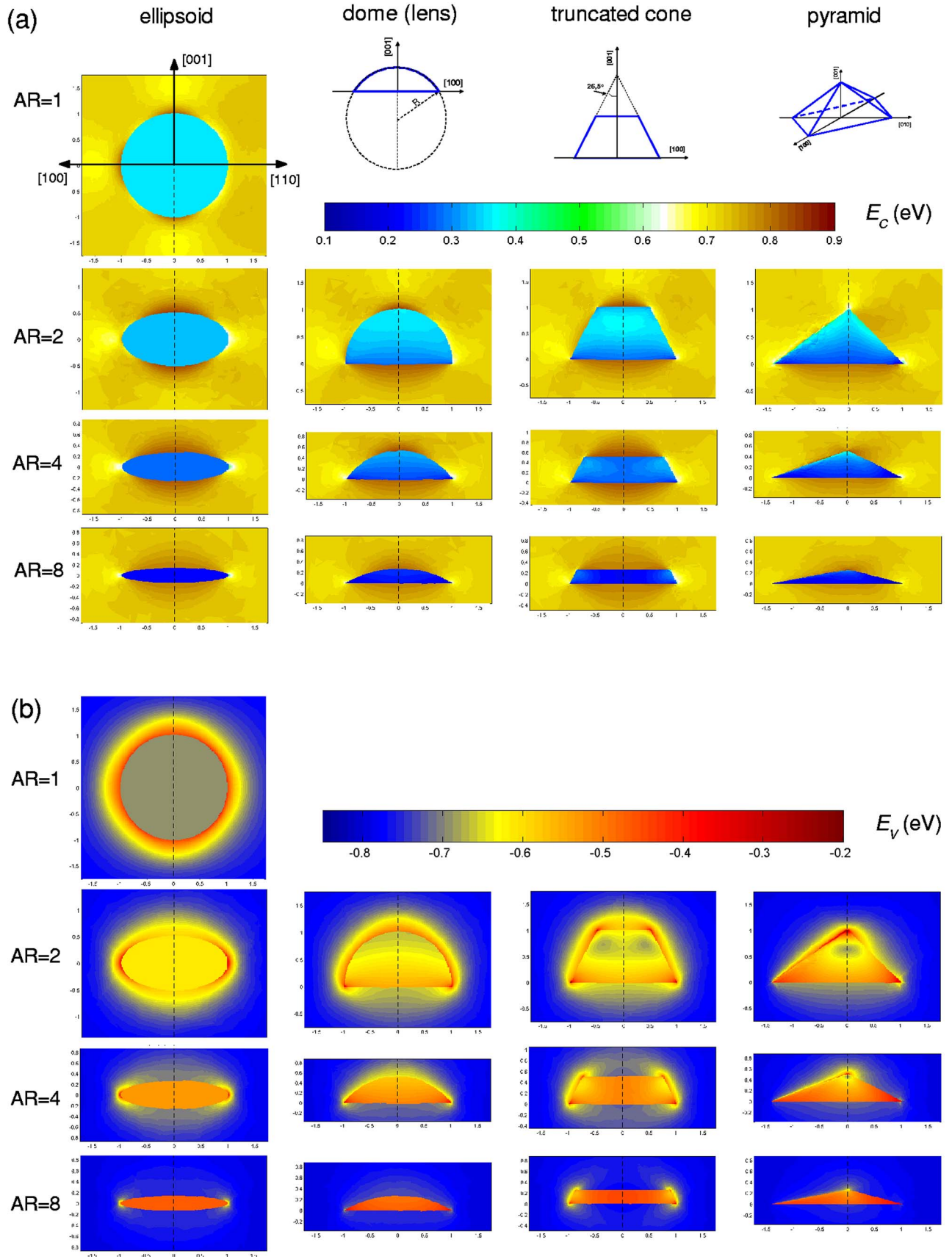


FIG. 3. (Color online) Band profile for several aspect ratios for isolated InAs/GaAs (001) QDs of different shapes. Each graph shows (100) and (-110) cross-section planes through the dot center on the left- and right-hand-side parts, respectively: (a) conduction band and (b) top valence band. Inserts illustrate the parental geometry details for the QDs.

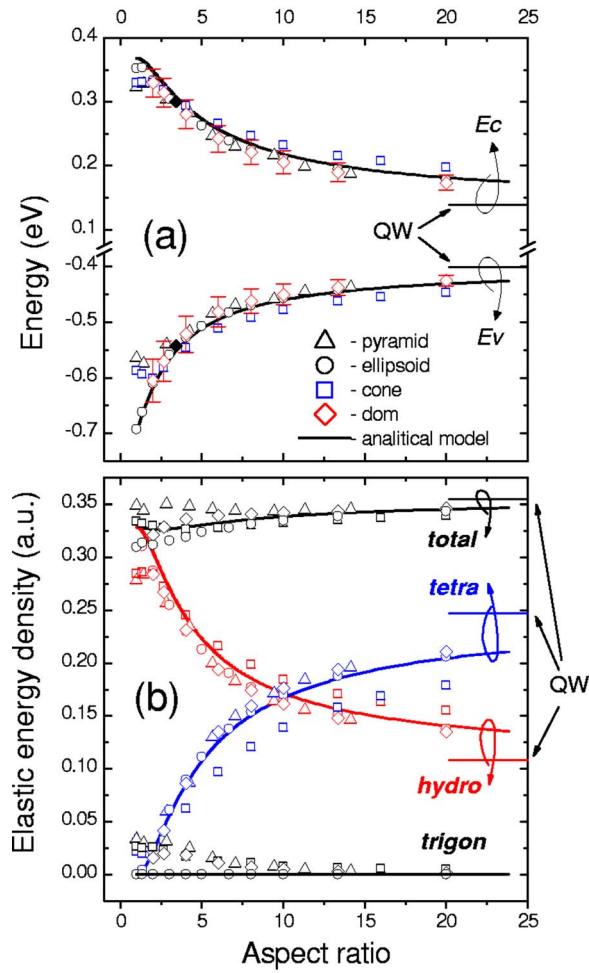


FIG. 4. (Color online) (a) Conduction and upper valence band edges within InAs/GaAs (001) QDs as a function of aspect ratio: open symbols—averaged numerical results for cubic elastic anisotropy approximation for different QD shapes: circles—ellipsoid, triangles—pyramidal, squares—cone, diamonds—dome, and filled diamond symbols show the results from Ref. 3. Solid lines present analytical results obtained within the IE model for the ellipsoidal shape. Short solid lines indicate band-edge positions for the InAs/GaAs (001) QW. The error bars show the mean deviation of energy values within the dome-shaped QD. (b) Elastic energy density within the InAs/GaAs (001) QD as a function of the aspect ratio. Symbols display averaged values of different components of the elastic energy obtained from numerical calculations. Solid lines represent the elastic energy density components for the ellipsoidal QD as obtained within the IE model. Short solid lines indicate values of the energy density components for an InAs/GaAs (001) quantum well.

5(c) provides a good representation of the localization potential for electrons, even despite the effects near the sharp edges of the dot.

For the top valence band, strain produces significant band bending in the matrix at low aspect ratios. This matches the nearly homogeneous valence band splitting for $AR=1$, shown in Fig. 1(c). Such a bending, together with “rounding up” of sharp corners inside the QD, provides a significant uplift of the top valence band within some “shell” around the QD [brighter areas, or lighter in Fig. 3(b)]. The shell generally includes both sides of the heterointerface.²² An exception is the ellipsoidal shape in which the shell includes the matrix side only. The valence band bending within this shell is of comparable magnitude for all shapes, as illustrated by

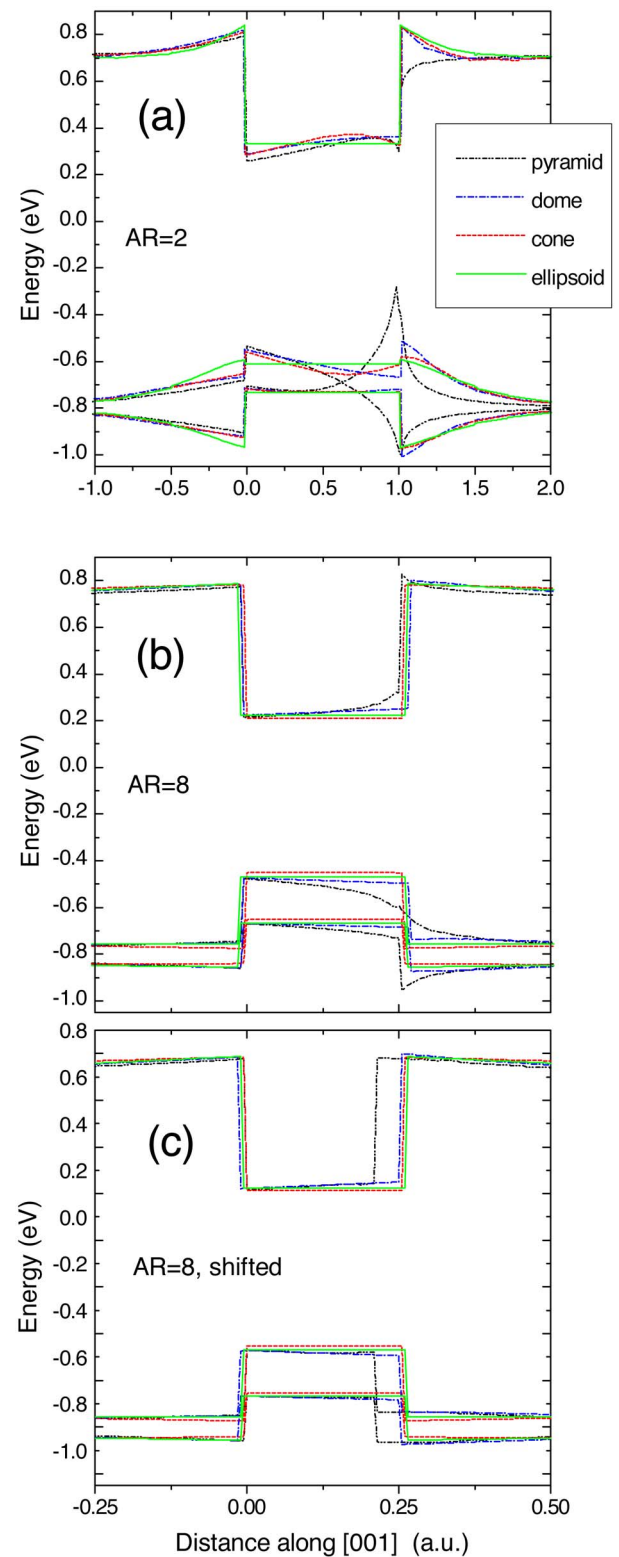


FIG. 5. (Color online) Band-edge profiles along the [001] direction through the dot center for different QD shapes: (a) aspect ratio=2; (b) aspect ratio=8; (c) aspect ratio=8 [same as (b)] along the [001] line shifted by 10% of the dot base size in the [110] direction.

1D band profiles in Fig. 5. Larger values are displayed at strain “concentrators” which are created at sharp corners of the QD. Note that the shell observed for a spherical QD, $AR=1$ in Fig. 3(b), has nearly spherical symmetry, to be compared to the strain distribution in Fig. 1(a).

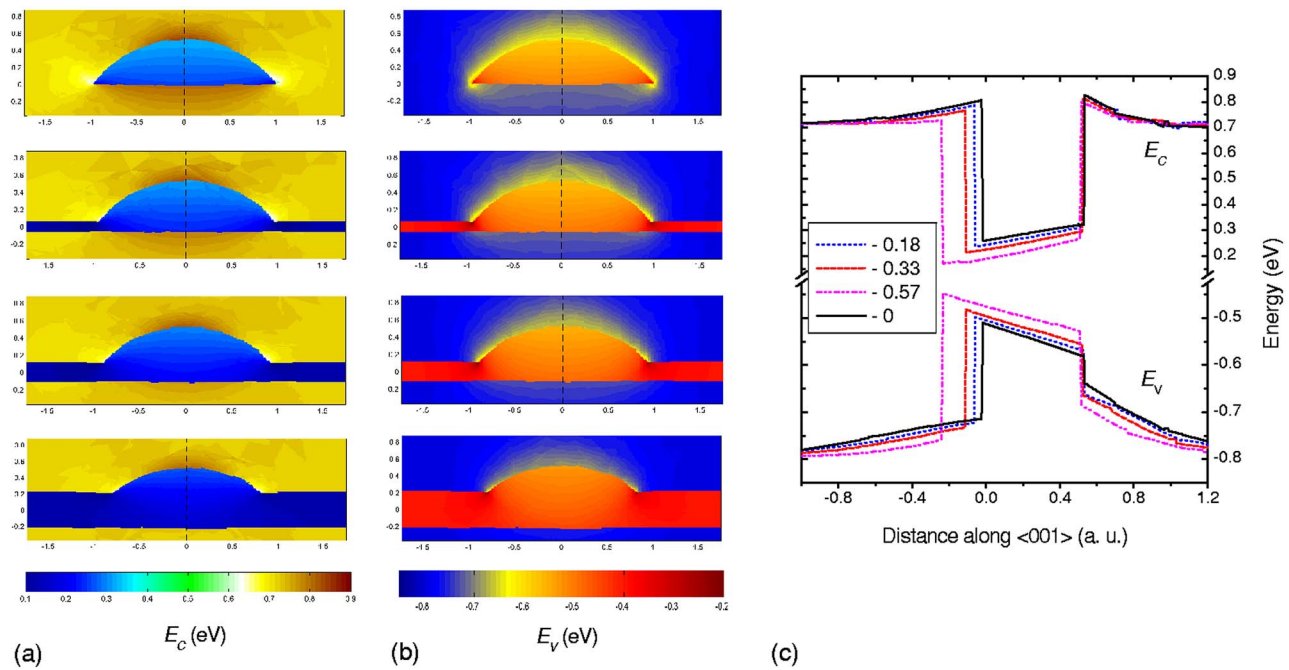


FIG. 6. (Color online) Confinement potential for lens-shaped InAs/GaAs (001) QD for different thicknesses of the wetting layer: (a) conduction band and (b) top valence band. Each graph shows (100) and $(\bar{1}10)$ cross-section planes through the dot center on the left- and right-hand-side parts, respectively. (c) Band-edge profile along the [001] direction through the dot center.

At larger aspect ratios, the valence band shell survives only in the form of “patches” along the sharp edges of the QD. This is observed for all shapes.

As a result, a representative valence band profile for flat dots along the [001] direction demonstrates a relatively weakly perturbed valence band edge at the matrix side of heterointerface, see Fig. 5(b) and 5(c). This common trend indicates that for large aspect ratios, the underlying strain variation is similar for all shapes and matches that for an ellipsoidal QD, [Fig. 1(a)]. Such an observation suggests that observation of the shell and its modification with the aspect ratio reflect some general (not shape-specific) features of strain distribution. These features may be sensitive only to the most basic material parameters, such as the ratio of elastic modulus in the dot and the matrix (assuming the same sign of misfit strain²³). Therefore, the shell feature may be expected to be common for a wide range of III-V material combinations (see Ref. 8 for a particular example).

There are important consequences of the shell feature for the hole localization potential in the QDs, which we would like to point out. Bending of the top valence band within the shell enlarges the effective QD size seen by localized holes and smoothes the sharpness of the confinement potential. Moreover, if the hole confinement potential in a QD is comparatively shallow, the shell provides a local energy minimum, in which the holes would tend to be localized. Actual localization is, of course, dependent on the QD size, which should not be too small.^{8,24} However, even if “full” localization is not possible, the “shell” can essentially redistribute the hole density within the QD, affecting the electron-hole overlap, the oscillator strength of the interband transition, or the sign of the quantum-confined Stark effect.²⁵ Effects from the shell may be particularly important in case of type-II band alignment with the electrons localized in the matrix.

C. Effect of wetting layer

The calculations above have not taken into account the wetting layer (WL), which is normally present in the SAQD heterostructures. We performed an additional series of calculations for the lens-shaped dots to include effects from the WL. The QDs were superimposed onto the WL to give varying relative thickness. [The relative thickness is defined as the ratio of the WL thickness to the overall thickness (height) of the whole structure, dot plus WL.] Figures 6(a) and 6(b) present two-dimensional (2D) cross sections of the potential profile for the conduction and upper valence bands, respectively. The 1D band profiles along the [001] line through the dot center are shown in Fig. 6(c). Results for a QD without the WL are also presented for comparison.

Variation in the band-edge positions with the WL thickness is represented by variation in brightness in Figs. 6(a) and 6(b). The results indicate that effects from the WL on the strain-modified band profile are negligible, provided that the relative WL thickness is small. This observation is in agreement with earlier results.²⁶ Quantitative estimates, available from Fig. 6(c), show that a WL of relative thickness ≈ 2 results in a shift of less than 20 meV. At larger relative thicknesses, the effect of the WL appears to be equivalent to that of increased aspect ratio. This can be qualitatively understood if one considers a WL as an extreme case of the QD.

If there is a region in which effects from the WL are noticeable, it is at the sharp in-plane edges of flat QDs. The WL affects the strain distribution in close proximity to them, as might be expected. The effect of the WL extends only over a distance comparable to its thickness. Effectively, the WL smoothes the sharp edges and therefore removes strain concentrators and the band profile perturbation, related to them. The effect from such a perturbation was not significant

anyway, because the in-plane direction provides the least contribution to the quantization energies. The wetting layer makes this effect even smaller.

IV. ELLIPSOIDAL MODEL

A. Formulation

Analysis performed in Sec. III B highlights the two most important results:

- (a) Major features of the strain-modified confinement potential, revealed for the ellipsoidal shape, can be applied to characterize the effective confinement potential for all other shapes which we considered.
- (b) Both the averaged confinement potential and its variation with the aspect ratio are similar for all shapes, reflecting similar trends in the strain redistribution.

From this we conclude that that

- (1) The aspect ratio is a governing parameter related to the QD shape,
- (2) Major features of the effective confinement potential can be obtained in an ellipsoidal approximation for the QD shape, at least for the (001) growth plane and except for very low aspect ratios.

The ellipsoidal approximation is very practical because a simple analytical solution of the elastic problem is available. The solution is based on the isotropic-elasticity approximation. We showed in Sec. III A that it is sufficient to accurately reproduce the most important features of the quantization potential.

One can question the validity of using averaged values of the band profile seen by electrons (or holes). We note that any variations in the potential profile within the dot happen on a very small spatial scale, because a quantum dot itself is small enough to enable quantum-mechanical confinement. Therefore, disregarding these variations is going to have little effect on the positions of size-quantized energy levels. Because of that, we suggest that the ellipsoidal model, which uses averaged localization potential, provides a good initial approximation for analysis of electronic spectra of QDs, with high predictive efficiency. Such an approximation is useful when the exact QD shape is not known and/or the most general features of the quantization potential are of the interest.

Obviously, the ellipsoidal model does have limitations. For example, it is unable to reveal any spatial potential deviations from the average value. These may affect the density distribution of the electron and/or hole wave functions and hence parameters sensitive to their overlap, such as the oscillator strength.

We have arrived at similar conclusions after the same analysis and calculations were performed for the InSb/GaSb (001) system (not presented here). In particular, the same general trends in strain variation with the aspect ratio were identified. Because of this, we would expect such trends to be common for a wide range of III-V materials, in view of the same symmetry of elastic modulus and deformation potentials. Thus, we suggest that the ellipsoidal approximation can be applied to the whole range of III-V zinc-blende

SAQD systems. Moreover, in addition to compressively strained QDs, the ellipsoidal model appears to be applicable also to tensile-strained SAQD systems. Indeed, in the linear elasticity approximation, strain is symmetric with respect to the sign of stress, as is the strain-induced band shift and/or splitting, at least to the first order of magnitude.

B. Growth plain orientation

Here, we address the question, to what extent the ellipsoidal model is applicable to orientations of the growth plane other than (001). Good accuracy of the isotropic elasticity approximation suggests that the effect of the aspect ratio on the quantization potential should be similar, at least qualitatively, for any crystallographic orientation. This assumption is supported by additional results, which are presented below.

We performed calculations for ellipsoidal InAs QDs grown on the (111) GaAs plane.²⁷ In this case, strain inside the QD is composed of hydrostatic and trigonal components. The total elastic energy density and its components are shown in Fig. 7(a) as a function of the aspect ratio. Here, results for isotropic and anisotropic elasticity approximations are presented by solid lines and filled circles, respectively. We find that the shear strain component increases with aspect ratio, in a similar way to the case of the (001) plane [Fig. 2(a)]. However, we find that it happens essentially not at the expense of the hydrostatic strain. A change in the hydrostatic energy density is not very big; instead, the total energy density inside the dot significantly increases with aspect ratio. This reflects further strain concentration inside the dot with the aspect ratio, i.e., redistribution of misfit strain from the matrix to the dot.

Strain-induced modification of the band-edge positions²⁸ is presented in Fig. 7(b). Qualitatively, it is similar to the (001) case shown in Fig. 2(b). However, absolute values of band shift and/or splitting are different. What is the reason for this? The hydrostatic deformation potentials and bulk modulus are isotropic by definition. For the shear deformation potentials, a condition for isotropy is $b\sqrt{3}/d=1$,^{18,29} which roughly holds for most of III-V zinc-blende materials. Therefore, variation of band shift and/or splitting with growth direction originates mainly from the elastic anisotropy which alters the magnitude of strain. This happens because shear elastic modulus μ_2 comes into play instead of μ_1 for the (111) orientation, as was discussed in Sec. II B. Typically, μ_2 is twice as large as μ_1 .

Note that still a good agreement is observed for the (111) plane between the isotropic and anisotropic elasticity approximations.

For arbitrary growth direction, the situation is expected to be intermediate between (001) and (111) cases, with the major effect in strain caused by an effective μ value. We suggest that the ellipsoidal model in the isotropic elasticity approximation still possesses good predictive ability, as long as an appropriate expression for the Lamé coefficients in terms of cubic elastic constants is used, in the spirit of Eq. (3). It has also been noted in earlier papers^{5,30–32} that the isotropic elasticity approximation provides a relatively accurate picture of strain effects in SAQDs.

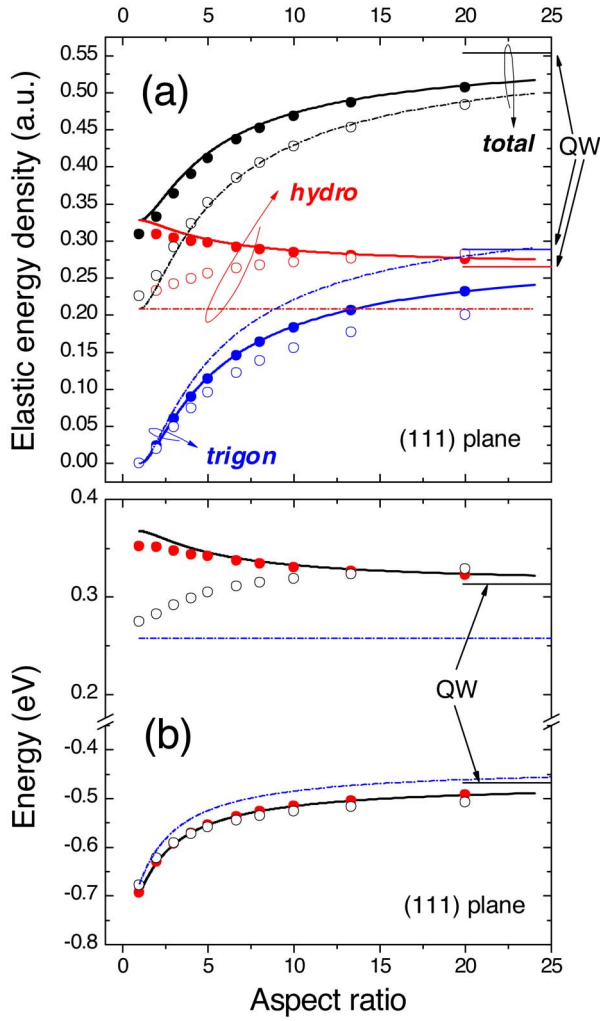


FIG. 7. (Color online) (a) Different elastic energy density components and (b) conduction and valence band-edge positions within the InAs/GaAs (111) ellipsoidal QD as a function of the aspect ratio. Solid lines—as obtained within the IE model; filled symbols—numerical results within cubic elastic anisotropy; dash-dotted lines and open symbols show results within the similar-elasticity approximation for isotropic and anisotropic elastic approximations, respectively. Short solid lines indicate band edges for InAs/GaAs (111) QW.

Such an observation may appear surprising in view of the fundamental elastic anisotropy of cubic III-V materials. The degree of anisotropy can be formally defined by deviation of $(c_{11}-c_{12})/2c_{44}$ from 1. For example, for GaAs $1-(c_{11}-c_{12})/2c_{44} \approx 0.55$. We suggest that a small effect of elastic anisotropy on the strain profile in III-V SAQDs is essential due to the coincidence of hard (soft) elastic directions in the dot and in the matrix. In that case, the normal force field produced by the lattice misfit at the heterointerface is sustained by a hard (soft) elastic direction in the dot against the hard (soft) direction in the matrix. Thus, anisotropy of individual moduli appears not to be as important as a *difference in anisotropy* between the dot and matrix materials. This difference is often very small. Indeed, the ratio of lattice stiffness along hard [111] and soft [001] directions, which is equal to μ_2/μ_1 , varies weakly between III-V zinc-blende materials.

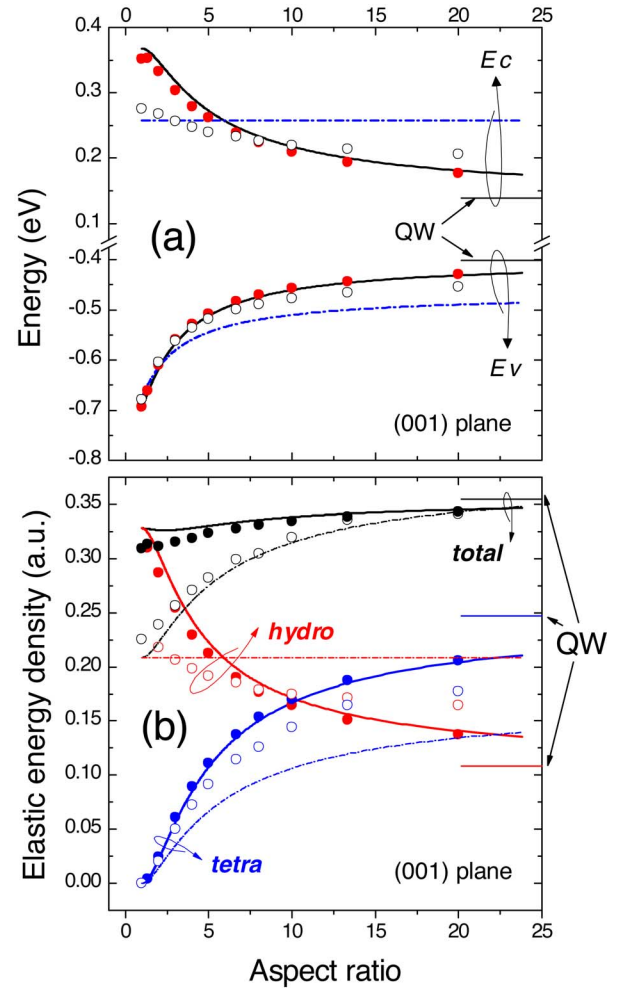


FIG. 8. (Color online) (a) Conduction and valence band-edge positions and (b) different elastic energy density components within the InAs/GaAs (001) ellipsoidal QD as a function of the aspect ratio. Filled symbols—numerical results within cubic elastic anisotropy; solid lines—as obtained within the IE model; dash-dotted lines and open symbols show results within the similar-elasticity approximation for isotropic and anisotropic elastic approximations, respectively. Short solid lines indicate band edges for InAs/GaAs (001) QW.

C. Comparison to the similar-elasticity approximation

Finally, we would like to compare our approach to the similar-elasticity (SE) approximation, in which the same elastic constants are assumed for dot and matrix materials. This approximation is typically employed in analytical calculations of the elastic problem for SAQDs.^{30–32} First, we note that for isotropic elasticity, the hydrostatic strain is constant and is equal to $e_m[(1+\nu)/(1-\nu)-3]$ in the dot and is zero in the matrix for *any* shape of the dot.¹¹ Hence, neither the conduction band energies in the dot nor in the matrix are at all affected by the dot shape. This reveals the most important limitation of the SE approximation. At the same time, variation in the shear strain and therefore in the position of the top valence band edge is similar to the case of different elastic constants. This is shown in Fig. 8; data for the SE approximation were obtained using the elastic constants of the GaAs matrix. However, there is a strong variation in the total strain energy density with the aspect ratio. This indicates that the origin of the increase in the valence band split-

ting is different. In this case it is due to further strain concentration in the dot, rather than strain redistribution between different components in the case of different elastic constants.

Combination of the similar-elasticity approximation with elastic anisotropy introduces some variation of the conduction band edge with the aspect ratio.³² The results of numerical calculations in such an approximation are presented by open circles in Figs. 7 and 8 for the (111) and (001) planes, respectively. One can see that for the conduction band variation as a function of aspect ratio, there is qualitative agreement between the two approximations in the case of the (001) plane. However, in case of the (111) plane, an *opposite* trend is observed. Such behavior reflects variation in the hydrostatic strain component, as depicted in Figs. 7(a) and 8(b).

The above comparison demonstrates that the SE approach provides a less accurate approximation as compared to isotropic elasticity, especially for the conduction band. In particular, it predicts a *qualitatively* different strain variation with aspect ratio in and around the dot.

V. SUMMARY

We have examined the effect of the quantum-dot shape on the strain-modified quantization potential. For this purpose, the band profile was calculated for the InAs/GaAs (100) system for a number of realistic dot shapes. The most important features of the quantization potential were analyzed, and their variation with the dot geometry was examined. The underlying variations in the characteristic strain components within the QD were revealed. As a result, the aspect ratio was suggested as a governing geometric parameter of the QD shape. This was used to justify approximation of the QD shape by an ellipsoid, which is expected to be valid for a wide range of III-V zinc-blende SAQD systems. Effects from incorporation of the wetting layer and from the growth plane orientation have been also analyzed.

For the ellipsoidal shape, reproducibility of the quantization potential was analyzed using different approximations for the elastic problem. It was shown that most essential features of the quantization potential are well reproduced in the isotropic elasticity approximation, provided that the effective elastic moduli are selected according to orientation of the growth plane. This result is particularly important because the isotropic approximation allows a simple analytical solution of the elastic problem for an ellipsoidal QD. On the other hand, it was shown that further simplification of the elastic problem (such as the similar-elasticity approximation) results in a strain profile which is different both qualitatively and quantitatively. As a result, it reproduces the quantization potential with significant distortion.

APPENDIX

Expressions below are valid for a coordinate system, in which the axes (x, y, z) are parallel to the principal axes (a, b, c) of the ellipsoid, respectively. The original¹¹ solution for an arbitrary ellipsoid is adopted for an oblate ellipsoid

($a=b>c$). We use the original notations for the strain components and constants. Usual tensor index notation and summation rules are employed.

The following notations are used for the scalar part f and deviatoric parts f_{ij} of the second-order tensor f_{ij} :

$$f = f_{mm} \quad \text{and} \quad f_{ij} = f_{ij} - \frac{1}{3}f\delta_{ij}.$$

1. DEFINITION OF MATERIAL PARAMETERS AND INTERMEDIATE COEFFICIENTS

Lattice mismatch strain (initial strain) e_{ij}^* ,

$$e_{11}^* = e_{22}^* = e_{33}^* = \frac{a_m - a_i}{a_i}, \quad e_{i \neq j}^* = 0,$$

where a_m and a_i are the lattice constants for the matrix and inclusion (QD) materials, respectively.

Elastic constants,

$$\text{In the matrix: } \kappa, \mu, \quad \sigma = \frac{3\kappa - 2\mu}{2(3\kappa + \mu)}.$$

In the inclusion: κ_i, μ_i .

Intermediate coefficients are

$$S_{1111} = S_{2222} = Qa^2I_{aa} + RI_a, \quad S_{2233} = Qc^2I_{ac} - RI_a,$$

$$S_{1122} = S_{2211} = Qa^2I_{ab} - RI_a, \quad S_{3333} = Qc^2I_{cc} + RI_c,$$

$$S_{1133} = Qc^2I_{ac} - RI_a, \quad S_{3311} = S_{3322} = Qa^2I_{ac} - RI_c,$$

$$S_{1212} = Qa^2I_{ab} + RI_a,$$

$$S_{1313} = S_{2323} = \frac{1}{2}Q(a^2 + c^2)I_{ac} + \frac{1}{2}R(I_a + I_c),$$

where

$$Q = \frac{3}{8\pi(1 - \sigma)}, \quad R = \frac{1 - 2\sigma}{8\pi(1 - \sigma)},$$

$$I_a = \frac{2\pi a^2 c}{(a^2 - c^2)^{3/2}} \left[\arccos\left(\frac{c}{a}\right) - \frac{c}{a} \left(1 - \frac{c^2}{a^2}\right)^{1/2} \right],$$

$$I_c = 4\pi - 2I_a, \quad I_{ac} = \frac{I_c - I_a}{3(a^2 - c^2)}, \quad I_{cc} = \frac{4\pi}{3c^2} - 2I_{ac},$$

$$I_{aa} = \frac{4\pi}{3a^2} - I_{ab} - I_{ac}, \quad I_{ab} = \frac{1}{3}I_{aa}.$$

2. STRAIN COMPONENTS INSIDE THE INCLUSION (QD)

For the orientation of the ellipsoid employed with the (001) growth plane, only diagonal components of the strain tensor e_{ij}^T are nonzero inside the QD,

$$e_{11}^T = e_{22}^T = -\frac{E}{D}e_{33}^T, \quad e_{33}^T = \frac{3e^*CD}{AE - BD},$$

where

$$A = 2 \frac{\mu}{\mu - \mu_i} + 6S_{3311} - 2 \frac{k}{k - k_i},$$

$$B = -2 \frac{\mu}{\mu - \mu_i} + 3S_{3333} - \frac{k}{k - k_i},$$

$$C = \frac{k}{k - k_i},$$

$$D = S_{1111} + S_{1122} - 2S_{3311} - \frac{\mu}{\mu - \mu_i},$$

$$E = S_{1133} - S_{3333} + \frac{\mu}{\mu - \mu_i}.$$

3. STRAIN IN THE MATRIX AT HETEROINTERFACE

The components of strain tensor $e_{ij}^C(\text{out})$ in the matrix just near the interface can be found from the values of $e_{ij}^C(\text{in})/e_{ij}^T$ at an adjacent point inside the inclusion by using the following expressions:¹¹

$$e^C(\text{out}) = e^C(\text{in}) - \frac{1}{3} \frac{1 + \sigma}{1 - \sigma} e^T - \frac{1 - 2\sigma}{1 - \sigma} e_{ij}^T n_i n_j,$$

$$\begin{aligned} e_{ij}^C(\text{out}) &= e_{ij}^C(\text{in}) + \frac{1}{1 - \sigma} e_{jk}^T n_j n_k n_i - e_{ik}^T n_k n_l - e_{lk}^T n_l n_i \\ &+ \frac{1 - 2\sigma}{3(1 - \sigma)} e_{jk}^T n_j n_k \delta_{il} - \frac{1}{3} \frac{1 + \sigma}{1 - \sigma} e^T \left(n_i n_l - \frac{1}{3} \delta_{il} \right), \end{aligned}$$

$$e_{il}^C = S_{ilmn} e_{mn}^T,$$

where n_i are components of the normal to the ellipsoid surface at the point (x, y, z) ,

$$n_1 = \frac{x}{a^2 h}, \quad n_2 = \frac{y}{b^2 h}, \quad n_3 = \frac{z}{c^2 h}, \quad h = \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right)^{1/2}.$$

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²²Note that the top valence band has different origin on different sides of heterointerface: it is more heavy-hole-like inside the QD and light-hole-like on the matrix side.

²³Note that for III-V SAQDs, the elastic modulus ratio and the sign of the misfit strain are correlated, because the materials with larger lattice constant are typically softer.

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